

You are managing an application that runs in Compute Engine. The application uses a custom HTTP server to expose an API that is accessed by other applications through an internal TCP/UDP load balancer. A firewall rule allows access to the API port from 0.0.0.0/0. You need to configure Cloud Logging to log each IP address that accesses the API by using the fewest number of steps. What should you do first?

- A. Enable Packet Mirroring on the VPC.
 - B. Install the Ops Agent on the Compute Engine instances.
 - C. Enable logging on the firewall rule.
 - D. Enable VPC Flow Logs on the subnet. **Most Voted**
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Your company runs an ecommerce website built with JVM-based applications and microservice architecture in Google Kubernetes Engine (GKE). The application load increases during the day and decreases during the night. Your operations team has configured the application to run enough Pods to handle the evening peak load. You want to automate scaling by only running enough Pods and nodes for the load. What should you do?

- A. Configure the Vertical Pod Autoscaler, but keep the node pool size static.
 - B. Configure the Vertical Pod Autoscaler, and enable the cluster autoscaler.
 - C. Configure the Horizontal Pod Autoscaler, but keep the node pool size static.
 - D. Configure the Horizontal Pod Autoscaler, and enable the cluster autoscaler. **Most Voted**
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A third-party application needs to have a service account key to work properly. When you try to export the key from your cloud project, you receive an error: "The organization policy constraint iam.disableServiceAccountKeyCreation is enforced." You need to make the third-party application work while following Google-recommended security practices.

What should you do?

- A. Enable the default service account key, and download the key.
- B. Remove the iam.disableServiceAccountKeyCreation policy at the organization level, and create a key.

- C. Disable the service account key creation policy at the project's folder, and download the default key.
 - D. Add a rule to set the iam.disableServiceAccountKeyCreation policy to off in your project, and create a key. **Most Voted**
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Your team is writing a postmortem after an incident on your external facing application. Your team wants to improve the postmortem policy to include triggers that indicate whether an incident requires a postmortem. Based on Site Reliability Engineering (SRE) practices, what triggers should be defined in the postmortem policy? (Choose two.)

- A. An external stakeholder asks for a postmortem
 - B. Data is lost due to an incident. **Most Voted**
 - C. An internal stakeholder requests a postmortem.
 - D. The monitoring system detects that one of the instances for your application has failed.
 - E. The CD pipeline detects an issue and rolls back a problematic release. **Most Voted**
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You are implementing a CI/CD pipeline for your application in your company's multi-cloud environment. Your application is deployed by using custom Compute Engine images and the equivalent in other cloud providers. You need to implement a solution that will enable you to build and deploy the images to your current environment and is adaptable to future changes. Which solution stack should you use?

- A. Cloud Build with Packer **Most Voted**
 - B. Cloud Build with Google Cloud Deploy
 - C. Google Kubernetes Engine with Google Cloud Deploy
 - D. Cloud Build with kpt
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You are creating a CI/CD pipeline in Cloud Build to build an application container image. The application code is stored in GitHub. Your company requires that production image builds are only run against the main branch and that the change control team approves all pushes to the main branch. You want the image build to be as automated as possible. What should you do? (Choose two.)

- A. Create a trigger on the Cloud Build job. Set the repository event setting to 'Pull request'.
 - B. Add the OWNERS file to the Included files filter on the trigger.
 - C. Create a trigger on the Cloud Build job. Set the repository event setting to 'Push to a branch' **Most Voted**
 - D. Configure a branch protection rule for the main branch on the repository. **Most Voted**
 - E. Enable the Approval option on the trigger.
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Your company is using HTTPS requests to trigger a public Cloud Run-hosted service accessible at the `https://booking-engine-abcdef.a.run.app` URL. You need to give developers the ability to test the latest revisions of the service before the service is exposed to customers. What should you do?

- A. Run the `gcloud run deploy booking-engine --no-traffic --tag dev` command. Use the `https://dev--booking-engine-abcdef.a.run.app` URL for testing. **Most Voted**
 - B. Run the `gcloud run services update-traffic booking-engine --to-revisions LATEST=1` command. Use the `https://booking-engine-abcdef.a.run.app` URL for testing.
 - C. Pass the `curl -H "Authorization:Bearer $(gcloud auth print-identity-token)"` auth token. Use the `https://booking-engine-abcdef.a.run.app` URL to test privately.
 - D. Grant the `roles/run.invoker` role to the developers testing the booking-engine service. Use the `https://booking-engine-abcdef.private.run.app` URL for testing.
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You are running a web application deployed to a Compute Engine managed instance group. Ops Agent is installed on all instances. You recently noticed suspicious activity from a specific IP address. You need to configure Cloud Monitoring to view the number of requests from that specific IP address with minimal operational overhead. What should you do?

- A. Configure the Ops Agent with a logging receiver. Create a logs-based metric.
- B. Create a script to scrape the web server log. Export the IP address request metrics to the Cloud Monitoring API. **Most Voted**

- C. Update the application to export the IP address request metrics to the Cloud Monitoring API.
 - D. Configure the Ops Agent with a metrics receiver.
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Your organization is using Helm to package containerized applications. Your applications reference both public and private charts. Your security team flagged that using a public Helm repository as a dependency is a risk. You want to manage all charts uniformly, with native access control and VPC Service Controls. What should you do?

- A. Store public and private charts in OCI format by using Artifact Registry. **Most Voted**
 - B. Store public and private charts by using GitHub Enterprise with Google Workspace as the identity provider.
 - C. Store public and private charts by using Git repository. Configure Cloud Build to synchronize contents of the repository into a Cloud Storage bucket. Connect Helm to the bucket by using `https://[bucket].storage.googleapis.com/[helmchart]` as the Helm repository.
 - D. Configure a Helm chart repository server to run in Google Kubernetes Engine (GKE) with Cloud Storage bucket as the storage backend.
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You use Terraform to manage an application deployed to a Google Cloud environment. The application runs on instances deployed by a managed instance group. The Terraform code is deployed by using a CI/CD pipeline. When you change the machine type on the instance template used by the managed instance group, the pipeline fails at the terraform apply stage with the following error message:

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Error waiting for Deleting Instance Template: The instance_template resource 'project/my-project/global/instanceTemplates/my-it-2022010101010100000000001' is already being used by 'projects/my-project/regions/us-central1/instanceGroupManagers/my-mig'
```

You need to update the instance template and minimize disruption to the application and the number of pipeline runs.

What should you do?

- A. Delete the managed instance group, and recreate it after updating the instance template.
 - B. Add a new instance template, update the managed instance group to use the new instance template, and delete the old instance template.
 - C. Remove the managed instance group from the Terraform state file, update the instance template, and reimport the managed instance group.
 - D. Set the create_before_destroy meta-argument to true in the lifecycle block on the instance template. **Most Voted**
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Your company operates in a highly regulated domain that requires you to store all organization logs for seven years. You want to minimize logging infrastructure complexity by using managed services. You need to avoid any future loss of log capture or stored logs due to misconfiguration or human error. What should you do?

- A. Use Cloud Logging to configure an aggregated sink at the organization level to export all logs into a BigQuery dataset.
 - B. Use Cloud Logging to configure an aggregated sink at the organization level to export all logs into Cloud Storage with a seven-year retention policy and Bucket Lock. **Most Voted**
 - C. Use Cloud Logging to configure an export sink at each project level to export all logs into a BigQuery dataset
 - D. Use Cloud Logging to configure an export sink at each project level to export all logs into Cloud Storage with a seven-year retention policy and Bucket Lock.
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You are building the CI/CD pipeline for an application deployed to Google Kubernetes Engine (GKE). The application is deployed by using a Kubernetes Deployment, Service, and Ingress. The application team asked you to deploy the application by using the blue/green deployment methodology. You need to implement the rollback actions. What should you do?

- A. Run the kubectl rollout undo command.
 - B. Delete the new container image, and delete the running Pods.
 - C. Update the Kubernetes Service to point to the previous Kubernetes Deployment. **Most Voted**
 - D. Scale the new Kubernetes Deployment to zero.
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You are building and running client applications in Cloud Run and Cloud Functions. Your client requires that all logs must be available for one year so that the client can import the logs into their logging service. You must minimize required code changes. What should you do?

- A. Update all images in Cloud Run and all functions in Cloud Functions to send logs to both Cloud Logging and the client's logging service. Ensure that all the ports required to send logs are open in the VPC firewall.
 - B. Create a Pub/Sub topic, subscription, and logging sink. Configure the logging sink to send all logs into the topic. Give your client access to the topic to retrieve the logs.
 - C. Create a storage bucket and appropriate VPC firewall rules. Update all images in Cloud Run and all functions in Cloud Functions to send logs to a file within the storage bucket.
 - D. Create a logs bucket and logging sink. Set the retention on the logs bucket to 365 days. Configure the logging sink to send logs to the bucket. Give your client access to the bucket to retrieve the logs. **Most Voted**
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You are building and running client applications in Cloud Run and Cloud Functions. Your client requires that all logs must be available for one year so that the client can import the logs into their logging service. You must minimize required code changes. What should you do?

- A. Deploy Falco or Twistlock on GKE to monitor for vulnerabilities on your running Pods.
 - B. Configure Identity and Access Management (IAM) policies to create a least privilege model on your GKE clusters.
 - C. Use Binary Authorization to attest images during your CI/CD pipeline. **Most Voted**
 - D. Enable Container Analysis in Artifact Registry, and check for common vulnerabilities and exposures (CVEs) in your container images.
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You have an application that runs in Google Kubernetes Engine (GKE). The application consists of several microservices that are deployed to GKE by using Deployments and Services. One of the microservices is experiencing an issue where a Pod returns 403 errors after the Pod has been running for more than five hours. Your development team is working on a solution, but the issue will not be resolved for a month. You need to ensure continued operations until the microservice is fixed. You want to follow Google-recommended practices and use the fewest number of steps. What should you do?

- A. Create a cron job to terminate any Pods that have been running for more than five hours.
 - B. Add a HTTP liveness probe to the microservice's deployment. **Most Voted**
 - C. Monitor the Pods, and terminate any Pods that have been running for more than five hours.
 - D. Configure an alert to notify you whenever a Pod returns 403 errors.
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You want to share a Cloud Monitoring custom dashboard with a partner team. What should you do?

- A. Provide the partner team with the dashboard URL to enable the partner team to create a copy of the dashboard. **Most Voted**
 - B. Export the metrics to BigQuery. Use Looker Studio to create a dashboard, and share the dashboard with the partner team.
 - C. Copy the Monitoring Query Language (MQL) query from the dashboard, and send the MQL query to the partner team.
 - D. Download the JSON definition of the dashboard, and send the JSON file to the partner team.
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You are building an application that runs on Cloud Run. The application needs to access a third-party API by using an API key. You need to determine a secure way to store and use the API key in your application by following Google-recommended practices. What should you do?

- A. Save the API key in Secret Manager as a secret. Reference the secret as an environment variable in the Cloud Run application. **Most Voted**
 - B. Save the API key in Secret Manager as a secret key. Mount the secret key under the /sys/api_key directory, and decrypt the key in the Cloud Run application.
 - C. Save the API key in Cloud Key Management Service (Cloud KMS) as a key. Reference the key as an environment variable in the Cloud Run application.
 - D. Encrypt the API key by using Cloud Key Management Service (Cloud KMS), and pass the key to Cloud Run as an environment variable. Decrypt and use the key in Cloud Run.
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Your company's security team needs to have read-only access to Data Access audit logs in the _Required bucket. You want to provide your security team with the necessary permissions following the principle of least privilege and Google-recommended practices. What should you do?

- A. Assign the roles/logging.viewer role to each member of the security team.
 - B. Assign the roles/logging.viewer role to a group with all the security team members.
 - C. Assign the roles/logging.privateLogViewer role to each member of the security team.
 - D. Assign the roles/logging.privateLogViewer role to a group with all the security team members. **Most Voted**
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Your team is building a service that performs compute-heavy processing on batches of data. The data is processed faster based on the speed and number of CPUs on the machine. These batches of data vary in size and may arrive at any time from multiple third-party sources. You need to ensure that third parties are able to upload their data securely. You want to minimize costs, while ensuring that the data is processed as quickly as possible. What should you do?

- A. Provide a secure file transfer protocol (SFTP) server on a Compute Engine instance so that third parties can upload batches of data, and provide appropriate credentials to the server.
Create a Cloud Function with a google.storage.object.finalize Cloud Storage trigger. Write code so that the function can scale up a Compute Engine autoscaling managed instance group
Use an image pre-loaded with the data processing software that terminates the instances when processing completes.
- B. Provide a Cloud Storage bucket so that third parties can upload batches of data, and provide appropriate Identity and Access Management (IAM) access to the bucket.
Use a standard Google Kubernetes Engine (GKE) cluster and maintain two services: one that processes the batches of data, and one that monitors Cloud Storage for new batches of data.
Stop the processing service when there are no batches of data to process.
- C. Provide a Cloud Storage bucket so that third parties can upload batches of data, and provide appropriate Identity and Access Management (IAM) access to the bucket.
Create a Cloud Function with a google.storage.object.finalize Cloud Storage trigger. Write code so that the function can scale up a Compute Engine autoscaling managed instance group.
Use an image pre-loaded with the data processing software that terminates the instances when processing completes. **Most Voted**
- D. Provide a Cloud Storage bucket so that third parties can upload batches of data, and provide appropriate Identity and Access Management (IAM) access to the bucket.

Use Cloud Monitoring to detect new batches of data in the bucket and trigger a Cloud Function that processes the data.
Set a Cloud Function to use the largest CPU possible to minimize the runtime of the processing.

You deployed an application into a large Standard Google Kubernetes Engine (GKE) cluster. The application is stateless and multiple pods run at the same time. Your application receives inconsistent traffic. You need to ensure that the user experience remains consistent regardless of changes in traffic and that the resource usage of the cluster is optimized.

What should you do?

- A. Configure a cron job to scale the deployment on a schedule
 - B. Configure a Horizontal Pod Autoscaler. **Most Voted**
 - C. Configure a Vertical Pod Autoscaler
 - D. Configure cluster autoscaling on the node pool.
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